SnapStudy AI

On-Device AI Study Assistant for Snapdragon-powered HP PCs

Project Proposal & Submission-Ready Concept

SnapStudy AI is a privacy-focused study assistant designed specifically with Snapdragon-powered HP PCs as the target deployment platform. It helps students understand their study material by generating summaries, answering questions, identifying key points, and creating practice MCQs.

Challenge alignment: The supplied Unstop reminder states that participants should build and optimize an AI-powered solution for Snapdragon-powered HP PCs and highlights the opportunity to showcase the innovation and receive a certificate of participation. ■filecite■turn0file0■L8-L18■

1. Problem Statement

Students often work with large PDFs, lecture notes, assignments, and technical documents. Finding important information, revising quickly, and creating practice questions can be time-consuming. Cloud-only AI tools may also require uploading private academic material and a reliable internet connection.

2. Proposed Solution

SnapStudy AI provides an AI-powered study workspace intended to run efficiently on a Snapdragon-powered HP PC. The application accepts study material, processes it, and provides four core capabilities: concise summaries, document-based question answering, key-point extraction, and automatic MCQ generation.

3. Why Snapdragon-powered HP PCs?

- **On-device AI target:** The AI inference layer is designed to be replaceable with a Snapdragon-optimized model/runtime.
- **Privacy:** Local processing can reduce the need to upload personal or academic documents to external servers.
- **Efficiency:** The architecture is designed to use hardware-accelerated AI where available rather than assuming CPU-only inference.
- **Offline-first direction:** Once a compatible local model is deployed, core study functions can be designed to work without continuous internet access.
- **Scalable deployment:** The user interface and document-processing layers remain independent from the model runtime, allowing Snapdragon-specific optimization without rebuilding the complete application.

4. System Architecture

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Student → PDF/Notes Upload → Document Parser → Text Chunking
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AI Inference Layer
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Summary Q&A; MCQs
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Student Dashboard

Target deployment:
Application → Optimized AI runtime/model → Snapdragon CPU/GPU/NPU
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5. Main Features

Feature	Purpose
Document upload	Accept PDF/lecture notes as study input.
AI summary	Produce a concise revision-oriented summary.
Ask the document	Answer questions using the supplied study material.
MCQ generator	Create practice questions with answers.
Key points	Extract important concepts and definitions.
Private/local AI target	Designed for local inference on the target PC.

6. Technology Stack

Layer	Proposed technology
Language	Python
User interface	Streamlit
Document processing	PyPDF / PDF text extraction
AI layer	Local/open-source language model, selected for Snapdragon deployment
Optimization target	Qualcomm Snapdragon AI ecosystem / compatible optimized runtime
Deployment target	Snapdragon-powered HP PC

7. Snapdragon Optimization Plan

- **Model selection:** Choose a compact language model suitable for local inference rather than relying on a large cloud-only model.
- **Model conversion/optimization:** Prepare the model for a Qualcomm-compatible execution path when Snapdragon hardware is available.
- **Hardware acceleration:** Target the Snapdragon AI compute resources, including the NPU where the selected model/runtime supports it.
- **Quantization:** Evaluate reduced-precision model variants to lower memory and compute requirements while maintaining acceptable answer quality.
- **Latency testing:** Measure document processing and response time on the target Snapdragon HP configuration.
- **Power efficiency:** Compare local inference cost and responsiveness with a CPU-only baseline.

8. Privacy & Accessibility

The intended deployment emphasizes local processing of study material. This can reduce exposure of private notes and assignments to external services. The interface is designed to be simple enough for students with basic computer skills and can provide value even when internet access is limited after the local model is installed.

9. Demonstration Flow

1. Open SnapStudy AI on the HP PC.
2. Upload a sample engineering/college PDF.
3. Generate a short summary.
4. Ask: “Explain the main concept in simple words.”
5. Generate five MCQs.
6. Show key points for quick revision.
7. Explain that the AI inference layer is designed for Snapdragon-specific optimization and local execution.

10. Important Prototype Disclosure

Hardware status: The current prototype should not claim that Snapdragon NPU performance has been tested unless the project is actually executed on compatible Snapdragon hardware. The correct submission language is that the solution is *designed/intended for optimization and deployment on Snapdragon-powered HP PCs*. Actual NPU benchmarking can be presented as a next validation step when compatible hardware is available.

11. Suggested Submission Description

SnapStudy AI is an on-device AI study assistant designed for Snapdragon-powered HP PCs. It transforms students’ PDFs and notes into concise summaries, document-grounded answers, key concepts, and practice MCQs. The application separates the user interface, document processing, and AI inference layers so that the model can be optimized for Snapdragon AI acceleration without redesigning the complete application. The solution prioritizes privacy by targeting local processing and is designed for efficient, accessible AI-assisted learning on Snapdragon-powered HP PCs.

12. Future Scope

- Snapdragon NPU benchmarking and latency measurement.
- Voice-based question answering.
- Handwritten-note recognition.
- Personalized revision schedules.
- Offline multilingual study support.
- Automatic flashcard generation.

Source basis: User-provided Unstop email for the Snapdragon AI Lab Build & Present Challenge. The email states that the participant should build and optimize an AI-powered solution for Snapdragon-powered HP PCs and lists the associated opportunities. ■filecite■turn0file0■L13-L18■